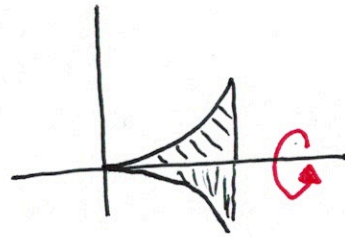
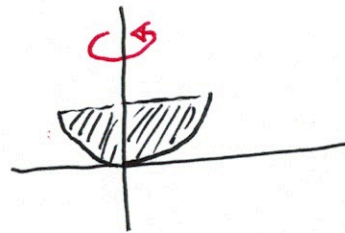


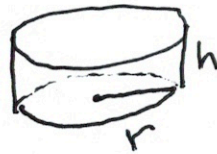
Section 6.2 Solids of revolution



rotate about
the x-axis



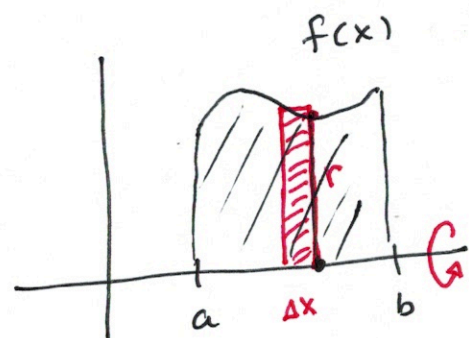
Rotate About
the y-axis



$$V = \pi \cdot r^2 \cdot h$$

Disk

Volume : Disk Method



$$\Delta x = \frac{b-a}{n}$$

rotate about
the x-axis



$$V = \pi \cdot r^2 \cdot h$$

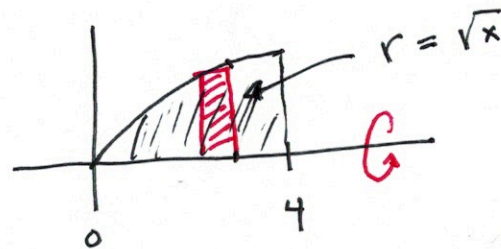
$$V = \pi \cdot [f(x)]^2 \cdot \Delta x$$

$$V = \int_a^b \pi \cdot [f(x)]^2 \cdot dx$$

NOTES

- 1) radius goes from axis of rotation to the function.
- 2) radius is perpendicular to axis of rotation.

Ex ① $y = \sqrt{x}$ on $[0, 4]$
rotate about the x -axis.



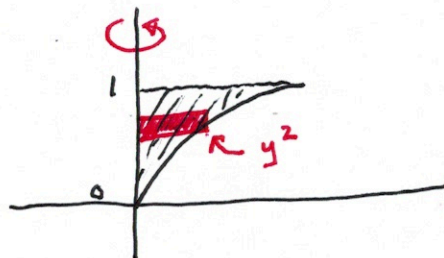
$$V = \int_0^4 \pi [\sqrt{x}]^2 \cdot dx$$

$$= \pi \cdot \frac{x^2}{2} \Big|_0^4$$

$$= \pi \cdot \frac{4^2}{2} - \pi \cdot \frac{0^2}{2}$$

$$= \underline{8\pi} \text{ units}^3$$

Ex ② $X = y^2$ on $[0, 1]$
rotate about y -axis.



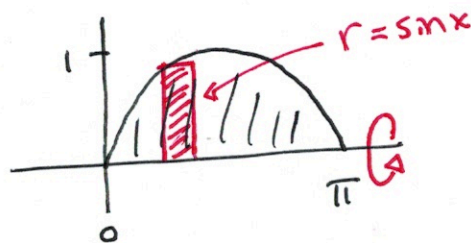
$$V = \int_0^1 \pi [y^2]^2 \cdot dy$$

$$V = \pi \cdot \frac{y^5}{5} \Big|_0^1$$

$$= \pi \cdot \frac{1}{5}$$

Ex ③ $y = \sin x$ on $[0, \pi]$

A) rotate about the x-axis.



$$V = \int_0^{\pi} \pi \cdot [\sin x]^2 \cdot dx$$

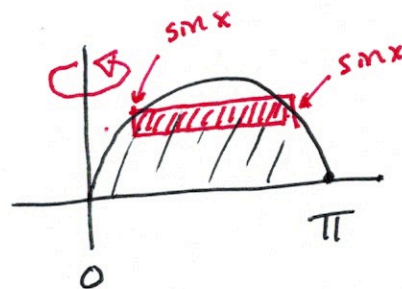
$$= \int_0^{\pi} \pi \cdot \left(\frac{1 - \cos 2x}{2} \right) dx$$

$$= \frac{\pi}{2} \int_0^{\pi} (1 - \cos 2x) dx$$

$$= \frac{\pi}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi}$$

$$= \frac{\pi}{2} [\pi] = \boxed{\frac{\pi^2}{2}}$$

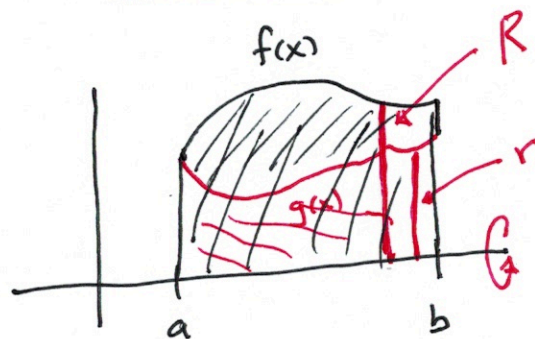
③ B)



Rotate About
the y -axis

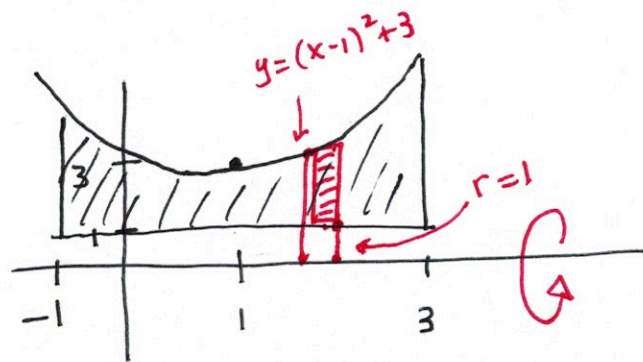
Not possible yet!

Washer Method



$$V = \int_a^b \pi [(\overset{R^2}{f(x)})^2 - (\overset{r^2}{g(x)})^2] dx$$

Ex (4) $y = (x-1)^2 + 3$, $y=1$, $x=-1$, $x=3$
Rotate About the x-axis. Set up integral.



Washer Method
 $\int \pi [R^2 - r^2] dx$

$$V = \int_{-1}^3 \pi [(x-1)^2 + 3]^2 - 1^2] dx$$